

The Pulmonary System

Ventilation, gas exchange, and the disorders that disrupt them — from V/Q mismatch and ABGs to obstructive vs restrictive disease, pediatric airways, and the bedside signs that follow.

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Module 10

Structure · Alterations · Pediatric

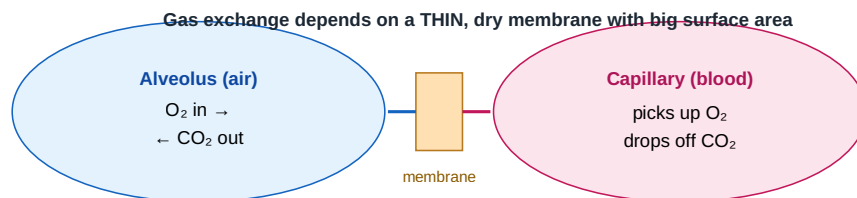
9 topics + cheat sheet

Structure & Function

SECTION 1

How air moves, how gases cross the membrane, and how breathing is controlled — the baseline every disorder disrupts.

Two jobs: **ventilation** (moving air in/out) and **gas exchange** (O_2 in, CO_2 out across the **alveolar–capillary membrane**). Both must work, and both must be **matched** — ventilation (air) needs perfusion (blood) right next to it.



Anything that **thickens** the membrane, **fills it with fluid**, or **reduces surface area** slows diffusion and causes hypoxemia.

What slows diffusion (→ hypoxemia)

Problem	What changes	Example
Thickened membrane	O_2 takes longer to cross	Pulmonary fibrosis
Fluid in the way	↑ diffusion distance	Pulmonary edema, pneumonia exudate
Lost surface area	Fewer working alveoli	Emphysema, atelectasis, lung compression
Low inspired O_2	↓ alveolar O_2 to begin with	High altitude (low FiO_2)

Key idea: the membrane works best when it is **thin, dry, and large**. Fibrosis thickens it; edema and pneumonia put **fluid** in the path (increasing diffusion distance); emphysema and atelectasis **destroy or collapse** surface area. All roads lead to **hypoxemia**.

PRACTICE

Q1. A patient with pulmonary fibrosis develops hypoxemia. Which mechanism explains it?

- A. Impaired diffusion across a thickened alveolar–capillary membrane
- B. Increased alveolar elasticity speeding O₂ diffusion
- C. Decreased membrane thickness
- D. Better ventilation overcoming the problem

A is correct. Fibrosis **thickens** and stiffens the membrane, slowing O₂ transfer into the blood. It reduces (not increases) elasticity, and ventilation can't overcome a diffusion barrier.

PRACTICE

Q2. Why does pulmonary edema reduce oxygenation?

- A. Alveolar fluid increases the diffusion distance for O₂
- B. Alveolar fluid shortens the diffusion distance
- C. Less interstitial fluid
- D. Increased surfactant

A is correct. Fluid in and around the alveoli lengthens the path O₂ must travel into the capillary, slowing diffusion → hypoxemia and dyspnea. (Pneumonia does the same with inflammatory **exudate**.)

The **respiratory center** in the brainstem sets the rate and depth of breathing, guided by chemoreceptors that sense the blood.

- **Central chemoreceptors** (brainstem) — mainly sense **CO₂/pH**; the strongest day-to-day driver of breathing.
- **Peripheral chemoreceptors** (carotid & aortic bodies) — sense **low O₂** (hypoxemia); they kick in to drive breathing when oxygen falls.

Key idea — the hypoxemia response: when arterial O₂ drops, the **peripheral chemoreceptors** fire, telling the brainstem to **increase rate and depth** → **hyperventilation / tachypnea**. It's a compensation to pull in more oxygen.

Pearl: tachypnea, nasal flaring, and accessory-muscle use are all signs of **increased respiratory drive / work of breathing** — the body trying to fix a gas-exchange problem. A *falling* rate in a tiring patient is ominous (drive is failing).

PRACTICE

Q1. A patient with hypoxemia begins to hyperventilate. Which mechanism explains this?

- A. Increased stimulation of peripheral chemoreceptors
- B. Decreased chemoreceptor stimulation
- C. Reduced chemoreceptor sensitivity
- D. Impaired chemoreceptor signaling

A is correct. Low arterial O₂ stimulates the peripheral chemoreceptors (carotid/aortic bodies), which signal the brainstem to raise rate and depth → hyperventilation. The other options would *blunt* the response.

PRACTICE

Q2. A patient at high altitude develops hypoxemia. Which mechanism explains it?

- A. Decreased inspired O_2 concentration lowering alveolar O_2
- B. Increased inspired O_2
- C. Increased pulmonary perfusion
- D. Impaired ventilation raising alveolar O_2

A is correct. At altitude the **fraction of inspired O_2 (FiO_2) and barometric pressure are lower**, so alveolar O_2 falls — the lungs are normal, but there's less oxygen to load. The body responds by hyperventilating.

The lungs are a major **acid–base** organ because **CO₂ is an acid**. Dissolved CO₂ + water → **carbonic acid**, which lowers pH. So how much you breathe directly sets your blood acidity.

The core relationship: Hypoventilation → CO₂ retained → more carbonic acid → pH falls = respiratory ACIDOSIS. Hyperventilation → CO₂ blown off → less acid → pH rises = respiratory ALKALOSIS.

Normal ABG values

Value	Normal range	Meaning
pH	7.35 – 7.45	<7.35 acidosis, >7.45 alkalosis
PaCO ₂	35 – 45 mmHg	The respiratory number (CO ₂ = acid)
PaO ₂	80 – 100 mmHg	Oxygenation; <80 = hypoxemia
HCO ₂₃	22 – 26 mEq/L	The metabolic (kidney) number

Pattern	pH	PaCO ₂	Cause
Respiratory acidosis	↓	↑	Hypoventilation (COPD, sedation, fatigue) retains CO ₂
Respiratory alkalosis	↑	↓	Hyperventilation (anxiety, early hypoxemia, pain) blows off CO ₂

Pearl — early vs late hypoxemia: early on, a hypoxemic patient **hyperventilates** → CO₂ drops → respiratory **alkalosis**. As they tire or the airway obstructs, CO₂ **climbs** → respiratory **acidosis** — a danger sign that the patient is failing.

PRACTICE

Q1. A patient with hypoventilation has an elevated PaCO₂. Which change drops the blood pH?

- A. Increased carbonic acid formation from retained CO_2
- B. Decreased carbonic acid from retained CO_2
- C. Increased carbonic acid from reduced CO_2
- D. Decreased carbonic acid from reduced CO_2

A is correct. Retained CO_2 combines with water to form more carbonic acid, lowering pH — respiratory acidosis.

PRACTICE

Q2. ABG: pH 7.32, PaCO_2 48, PaO_2 60. Which interpretation fits?

- A. Reduced alveolar ventilation causing hypoxemia and respiratory acidosis
- B. Excess ventilation causing hypocapnia and respiratory alkalosis
- C. Isolated diffusion problem with normal CO_2
- D. Metabolic acidosis with normal CO_2

A is correct. Low pH + **high PaCO_2** = respiratory acidosis from hypoventilation, and the low PaO_2 shows hypoxemia. (This is Elena's stage-3 ABG in the case study — her lung was being compressed by pleural fluid.)

SECTION 2

Alterations of Pulmonary Function

Obstructive vs restrictive disease, the mechanisms of hypoxemia, alveolar/pleural disorders, and the systemic signs that follow.

The master split. **Obstructive** = trouble getting air **out** (high resistance).
Restrictive = trouble getting air **in** (low compliance/stiff lungs).

	Obstructive	Restrictive
Problem	↑ airway resistance — can't exhale fully	↓ lung compliance — can't expand fully
Examples	Asthma, COPD (chronic bronchitis, emphysema), bronchiolitis	Pulmonary fibrosis, ARDS, atelectasis, pleural effusion, chest-wall problems
Hallmarks	Wheeze, prolonged expiration , air trapping, barrel chest, pursed-lip breathing	Rapid, shallow breaths (tachypnea), ↓ lung volumes
Airflow	Air gets in easier than it gets out → air trapping	Lungs resist filling; breaths are small

Asthma vs emphysema — the obstruction mechanism

- **Asthma:** reversible **bronchial smooth-muscle constriction** (+ inflammation, mucus) narrows the lumen → wheeze and prolonged expiration.
- **Emphysema (COPD):** **destruction of alveolar walls** → loss of elastic recoil → small airways collapse on exhalation → **air trapping**, barrel chest, pursed-lip breathing.

Key idea: obstructive disease raises **resistance** (the problem is **exhaling** — prolonged expiration, air trapping). Restrictive disease lowers **compliance** (the problem is **inhaling** — stiff lungs force rapid, shallow breaths). Severe obstruction traps CO₂ → **hypercapnia**.

PRACTICE

Q1. A 70-year-old with a 50-pack-year history has a barrel chest, pursed-lip breathing, and prolonged expiration. Which process most directly causes the airflow limitation?

- A. Destruction of alveolar walls and narrowing/collapse of small airways
- B. Increased elastic recoil improving expiration
- C. Excess surfactant
- D. Acute reversible bronchospasm only

A is correct. This is emphysema/COPD (Marcus in the case study). Alveolar-wall destruction **reduces elastic recoil**, so airways collapse on exhalation → air trapping and chronic obstruction. COPD is chronic, not a single acute bronchospasm.

PRACTICE

Q2. A patient with restrictive lung disease develops tachypnea. Which mechanism explains it?

- A. Decreased lung compliance forcing rapid, shallow breaths
- B. Increased lung compliance
- C. Reduced airway resistance
- D. Improved gas exchange

A is correct. Stiff (low-compliance) lungs are hard to expand, so the patient compensates with fast, shallow breathing to move enough air.

PRACTICE

Q3. A patient with asthma has wheezing and prolonged expiration. The main mechanism is:

- A. Bronchial smooth-muscle constriction narrowing the lumen
- B. Decreased airway inflammation
- C. Airway widening
- D. Increased compliance

A is correct. Bronchoconstriction (with inflammation and mucus) raises airway resistance, producing wheeze and a prolonged expiratory phase.



Hypoxemia = low arterial O_2 . Gas exchange needs **ventilation (V)** and **perfusion (Q)** matched. When they're out of balance, oxygen suffers.

Mechanism	What happens	Classic example
V/Q mismatch	Air & blood not matched in the same area	Most lung disease
Dead space (V without Q)	Alveoli ventilated but not perfused	Pulmonary embolism (blocked perfusion)
Shunt (Q without V)	Blood passes unventilated alveoli	Pneumonia, atelectasis, edema (filled/collapsed alveoli)
Diffusion defect	Thick/fluid-filled membrane	Fibrosis, edema
Low inspired O_2	Less O_2 to begin with	High altitude
Hypoventilation	Not enough air moved	Sedation, COPD, neuromuscular weakness

Key idea — pulmonary embolism: a clot blocks **perfusion** to a lung region. Those alveoli are still ventilated but get **no blood flow** (dead space) → V/Q mismatch → hypoxemia. The problem is **perfusion, not ventilation** — a favorite exam point.

Pearl: chronic hypoxemia triggers **hypoxic pulmonary vasoconstriction** — vessels in low-oxygen lung clamp down to redirect blood. Over the whole lung (e.g., COPD), this raises pulmonary pressures → **pulmonary hypertension** (Topic 7).

PRACTICE

Q1. A patient develops a pulmonary embolism with hypoxemia. Which mechanism explains it?

A. Decreased perfusion causing ventilation–perfusion mismatch

- B. Increased perfusion causing mismatch
- C. Decreased ventilation as the primary cause
- D. Uniform perfusion causing mismatch

A is correct. The embolus blocks blood flow, so ventilated alveoli aren't perfused (dead space) → V/Q mismatch → hypoxemia. PE primarily affects **perfusion**.

PRACTICE

Q2. A patient with chronic hypoxemia develops pulmonary hypertension. Which mechanism explains it?

- A. Increased pulmonary vasoconstriction
- B. Decreased pulmonary vasoconstriction
- C. Increased pulmonary blood flow
- D. Reduced pulmonary vascular resistance

A is correct. Sustained low O_2 causes widespread hypoxic pulmonary vasoconstriction, raising pulmonary arterial pressure over time.

These conditions fill, collapse, or compress the alveoli — cutting surface area and oxygenation.

Disorder	What happens	Result
Pulmonary edema	Fluid floods alveoli (often from left heart failure or capillary injury)	↑ diffusion distance → dyspnea, crackles, hypoxemia
ARDS	Diffuse alveolar injury → ↑ capillary permeability → protein-rich fluid floods alveoli	Severe refractory hypoxemia; triggers: sepsis, aspiration , toxic gas, trauma
Atelectasis	Alveolar collapse (obstruction, compression, or surfactant loss)	↓ ventilation in that area → shunt → hypoxemia
Pneumonia	Infection fills alveoli with inflammatory exudate	↑ diffusion distance → hypoxemia, productive cough, fever
Pleural effusion / empyema	Fluid (or infected pus = empyema) in the pleural space compresses lung	Restricts expansion → decreased breath sounds, dyspnea

Key idea — ARDS: the injury (sepsis, aspiration of gastric contents, toxic gas) makes the alveolar–capillary membrane **leaky** (increased permeability). Protein-rich fluid floods the alveoli, causing diffuse injury, stiff lungs, and severe hypoxemia that's hard to reverse with oxygen alone.

Pearl — pleuritic pain: sharp chest pain that **worsens with deep breaths** comes from inflammation of the **parietal pleura** (it has pain fibers; the lung itself doesn't). An infected effusion that becomes thick and purulent is an **empyema** — bacteria seed the pleural space with exudate and fibrin (Elena's case).

PRACTICE

Q1. After aspirating gastric contents, a patient develops dyspnea, crackles, and diffuse alveolar injury. Which mechanism best explains it?

- A. Increased inflammatory response injuring the alveolar–capillary membrane
- B. Decreased inflammation
- C. Reduced membrane defense
- D. Impaired alveolar repair

A is correct. Aspiration triggers intense inflammation that damages the membrane and floods alveoli with fluid — the ARDS pattern.

PRACTICE

Q2. A 64-year-old with pneumonia develops sharp left chest pain worse with inspiration; CXR shows a large pleural fluid collection compressing the lung. What explains the pain?

- A. Inflammation of the parietal pleura (pleuritic pain)
- B. Airway obstruction with hyperinflation
- C. Surfactant deficiency causing collapse
- D. Increased pulmonary blood flow

A is correct. Pleural inflammation stimulates pain fibers in the parietal pleura, which stretch during lung expansion — sharp pain that worsens with deep breaths (Elena, Stage 1).

PRACTICE

Q3 (SATA). Which reflect worsening alveolar gas exchange when pleural fluid compresses the lung? (a) reduced alveolar ventilation from restricted expansion, (b) less alveolar surface area, (c) V/Q mismatch in compressed regions, (d) increased diffusion distance from fluid compression.

a, b, c, and d are all correct. Compression restricts expansion (less ventilation), reduces working surface area, mismatches V/Q, and adds a fluid barrier — together driving hypoxemia.

Chronic lung disease shows up across the body. These are the recognizable downstream signs.

Sign	Mechanism
Cyanosis	↑ deoxygenated hemoglobin in capillary blood → bluish skin/ mucous membranes
Clubbing	Chronic hypoxemia → growth factors stimulate tissue/ vascular proliferation in the fingertips
Pulmonary hypertension	Chronic hypoxic vasoconstriction → high pulmonary artery pressure
Cor pulmonale	The right ventricle fails against chronic pulmonary hypertension → JVD, edema, hepatomegaly

Key idea: cyanosis is about the **amount of deoxygenated hemoglobin**, not just saturation — it appears when deoxy-Hb rises enough to look blue. **Clubbing** is a *chronic* hypoxemia sign (growth-factor–driven tissue growth), so it points to long-standing disease.

PRACTICE

Q1. A patient with hypoxemia develops cyanosis. Which mechanism explains it?

- A. Increased deoxygenated hemoglobin
- B. Decreased deoxygenated hemoglobin
- C. Increased oxygen saturation
- D. Reduced hemoglobin concentration

A is correct. Cyanosis appears when the amount of deoxygenated hemoglobin rises enough to give a bluish color. Higher saturation would reduce it.

PRACTICE

Q2. A patient with chronic hypoxemia develops finger clubbing. Which mechanism explains it?

- A. Increased growth factors causing distal tissue proliferation
- B. Decreased growth factors causing tissue loss
- C. Reduced tissue growth
- D. Impaired growth signals

A is correct. Chronic low oxygen drives growth-factor–mediated vascular and tissue proliferation at the fingertips, producing clubbing.

SECTION 3

Pediatric & Clinical Application

Why children's lungs fail differently, the neonatal and pediatric airway emergencies, and the environmental triggers in the headlines.

Surfactant lowers surface tension and keeps alveoli open at the end of each breath. Premature infants haven't made enough — the core of **neonatal respiratory distress syndrome (RDS)**.

Key idea — RDS: immature type II cells → **surfactant deficiency** → **high surface tension** → alveoli **collapse** on exhalation (atelectasis) → impaired gas exchange and hypoxemia. More surfactant would *prevent* collapse, so the answer is always "decreased surfactant → increased surface tension."

Signs of increased work of breathing (any age, vivid in infants)

- **Retractions** — greater inspiratory effort makes intrathoracic pressure **more negative**; a child's **compliant chest wall** gets pulled inward.
- **Nasal flaring** and **accessory-muscle use** — recruiting extra muscles to move air.
- **Tachypnea** — faster breathing to compensate for hypoxemia.
- Premature infants may also **hypoventilate** (immature drive/muscles) → CO₂ retention (hypercapnia).

Pearl — why kids show retractions: children have a **more compliant (softer) chest wall** than adults. When they work hard to inhale against an obstruction, the soft tissues between/under the ribs suck inward. *Stiffer* chest walls would be *less* likely to retract.

PRACTICE

Q1. A premature infant develops RDS with alveolar collapse. Which mechanism explains it?

- A. Impaired surfactant production increasing surface tension
- B. Increased surfactant decreasing surface tension
- C. Decreased surfactant decreasing surface tension
- D. Surfactant excess

A is correct. Too little surfactant raises surface tension, so alveoli collapse on exhalation (atelectasis), impairing gas exchange.

PRACTICE

Q2. An infant with airway obstruction develops chest retractions. Which mechanism explains it?

- A. Increased negative intrathoracic pressure pulling the compliant chest wall inward
- B. Decreased negative intrathoracic pressure pushing it outward
- C. Increased positive intrathoracic pressure
- D. A stiff, non-compliant chest wall

A is correct. Harder inspiratory effort against obstruction makes intrathoracic pressure more negative; the child's compliant chest wall is drawn inward → retractions.

Children's airways are **smaller**, so the same swelling causes far more obstruction. Know the classic patterns.

Condition	Where / cause	Hallmark
Croup	Upper airway — subglottic inflammation/edema (viral)	Barking cough + inspiratory stridor , hoarse voice
Bronchiolitis (RSV)	Lower small airways — inflammation, edema, mucus	Wheeze, air trapping (can't fully exhale), respiratory distress in infants
Foreign-body aspiration	Object lodges in an airway	Complete block → trapped air absorbs → atelectasis ; partial block → air trapping
Pneumonia	Alveoli fill with exudate	Hypoxemia, fever, focal findings

Why the airway sounds differ

Stridor (harsh, on inspiration) = **upper** airway narrowing → turbulent airflow (croup). **Wheeze** (on expiration) = **lower** airway narrowing (bronchiolitis, asthma). Narrowing raises resistance and makes airflow turbulent and noisy.

Environmental & "current events" triggers

- **Air pollution / urban air** → airway **inflammation** and bronchial hyperresponsiveness → more asthma exacerbations (bronchoconstriction).
- **Wildfire smoke** → airway irritation → tachypnea (raised respiratory drive).
- **Viral outbreaks** → more pediatric pneumonia/bronchiolitis → alveolar exudate or small-airway obstruction → hypoxemia.
- **Rising prematurity** → more surfactant-deficiency RDS.

Key idea — air trapping in bronchiolitis: inflamed, narrowed small airways let air **in** more easily than **out**, so air gets trapped distally (hyperinflation). The unifying mechanism is **airway obstruction preventing full expiration**.

Q1. A 6-year-old has a barking cough and inspiratory stridor that worsened overnight, with a hoarse voice. What is the primary mechanism?

- A. Subglottic airway inflammation and edema narrowing the upper airway
- B. Surfactant deficiency in the lower airways
- C. Alveolar fluid impairing oxygen exchange
- D. Bronchial smooth-muscle constriction limiting expiration

A is correct. This is croup (Noah's case). Subglottic edema narrows the **upper** airway → turbulent inspiratory airflow = stridor and a barking cough. Bronchospasm causes *wheeze*, not stridor.

PRACTICE

Q2. A child aspirates a foreign body and develops atelectasis. Which mechanism explains it?

- A. Complete obstruction → trapped distal air is absorbed → alveolar collapse
- B. Partial obstruction causing hyperinflation only
- C. Open airway maintaining ventilation
- D. Widened airway increasing ventilation

A is correct. When an object fully blocks an airway, the air beyond it is gradually absorbed into the blood, and the unventilated alveoli collapse (resorption atelectasis).

PRACTICE

Q3. Urban air pollution increases a child's asthma exacerbations. The airflow limitation is mainly from:

- A. Bronchial smooth-muscle constriction narrowing the airway
- B. Smooth-muscle relaxation widening the airway
- C. Pleural stiffening

D. Reduced diaphragm contraction

A is correct. Pollutants irritate the airways and trigger hyperresponsiveness → bronchoconstriction, which raises resistance and limits airflow.

□ EXAM

Pulmonary Cheat Sheet

The highest-yield patterns — review the night before.

ABG one-liners

Pattern	pH / PaCO ₂	Cause
Respiratory acidosis	↓ pH, ↑ CO ₂	Hypoventilation (COPD, sedation, tiring patient)
Respiratory alkalosis	↑ pH, ↓ CO ₂	Hyperventilation (anxiety, pain, early hypoxemia)
Normal	7.35–7.45 / 35–45	PaO ₂ 80–100; <80 = hypoxemia

Don't-miss facts

- **CO₂ is acid:** retain CO₂ → acidosis; blow off CO₂ → alkalosis.
- **Hypoxemia** → **peripheral chemoreceptors fire** → **hyperventilation/tachypnea**.
- **Obstructive** = ↑ resistance, can't exhale (asthma, COPD; wheeze, prolonged expiration, air trapping). **Restrictive** = ↓ compliance, can't inhale (fibrosis, ARDS; rapid shallow breaths).
- **Emphysema:** alveolar-wall destruction → lost recoil → air trapping, barrel chest, pursed-lip breathing.
- **PE** = blocked **perfusion** → dead space → V/Q mismatch → hypoxemia.
- **Pneumonia/edema** = fluid/exudate → ↑ diffusion distance. **Atelectasis** = collapse → shunt.

- **ARDS** = ↑ capillary permeability (sepsis, aspiration, toxic gas) → alveoli flood → severe hypoxemia.
- **Pleuritic pain** = parietal pleura inflammation; **empyema** = infected (pus) pleural fluid.
- **Cyanosis** = ↑ deoxygenated Hb. **Clubbing** = chronic hypoxemia (growth factors).
- **Chronic hypoxemia** → **hypoxic vasoconstriction** → **pulmonary HTN** → **cor pulmonale** (right heart failure).
- **RDS** (premature) = ↓ surfactant → ↑ surface tension → collapse. **Croup** = stridor (upper). **Bronchiolitis** = wheeze + air trapping (lower).
Retractions = compliant chest wall + ↑ work of breathing.

□ TERMS

Glossary

Quick definitions of the Week 10 vocabulary.

Ventilation

Movement of air in and out of the lungs.

Perfusion

Blood flow through the pulmonary capillaries.

V/Q mismatch

Imbalance between ventilation and perfusion; the most common cause of hypoxemia.

Dead space

Ventilated alveoli that are **not perfused** (e.g., pulmonary embolism).

Shunt

Perfused alveoli that are **not ventilated** (e.g., pneumonia, atelectasis).

Hypoxemia / Hypercapnia

Low arterial O₂ / high arterial CO₂.

Compliance

How easily the lungs stretch; **low** in restrictive disease.

Airway resistance

Opposition to airflow; **high** in obstructive disease.

Surfactant

Substance that lowers alveolar surface tension and prevents collapse; deficient in neonatal RDS.

Atelectasis

Collapse of alveoli → reduced ventilation and hypoxemia.

ARDS

Acute respiratory distress syndrome — diffuse alveolar injury with increased capillary permeability and severe hypoxemia.

Pleural effusion / Empyema

Fluid in the pleural space / infected (purulent) pleural fluid.

Stridor

Harsh inspiratory sound from **upper** airway narrowing (e.g., croup).

Cor pulmonale

Right-heart failure caused by chronic pulmonary hypertension.

Clubbing

Bulbous fingertip change from chronic hypoxemia.